WHONDRS HJW Sediment and Surface Water Sampling Protocol

This protocol will describe sediment and river water sampling (~1.5hrs) for the RC Workshop.

MATERIALS

In addition to the WHONDRS sampling kit, you will need the following:	**The WHONDRS sampling kit should contain the following:**
1. Ideally two people to do the sampling 2. Cooler with wet or blue ice for keeping samples cold in the field 3. Method for collecting latitude and longitude in decimal degrees in the field (smart phone is sufficient) 4. Method for taking pictures of the field site (smart phone is sufficient) 5. -20 C freezer for freezing the ice packs in advance of shipping (<u>do not freeze the samples</u> but be sure to freeze the ice packs at least 48 hours prior to shipping) 6. Refrigerator to store samples prior to shipping 7. Access to a FedEx office for shipping samples back overnight (WHONDRS pays for shipping) 8. Thermometer to measure temperature in the water column 9. Safety glasses (<u>some vials are pre-acidified and should be treated with care</u>) 10. Metric measuring tape for estimating water depth in cm 11. If needed, use a rock, brick, stake, etc. to tie the dissolved oxygen sensor to, so it stays in place	1. Hard copy metadata form 2. Nitrile gloves to minimize contamination 3. One sterile/clean sediment scoops 4. One foil-wrapped metal scoopula to assist in distributing sediment from the sediment scoop 5. One 50mL syringe 6. Three proteomic friendly conical tubes for sieved sediments. 7. Three **pre-acidified** 60mL amber vials for filtered water 8. Three 40mL amber vials for filtered water 9. Sterivex filter and a spare one in case the main one clogs 10. 2mm sieve with a catch pan

IDENTIFY SAMPLING LOCATIONS

After confirming that you have all of the needed gear, identify a wadeable (i.e., **in water**) sampling location for sediment collection in a depositional zone
Sediments will be collected from one stream reach, and potentially a single depositional zone if there is sufficient fine-grained sediment. Once sampling site is identified, collect metadata, conduct water sampling, and then collect sediments (**collect water before sediments**).

COLLECT METADATA ON THE PROVIDED METADATA SHEET

1. Record your name, the stream name, and the sample ID. The sample ID is the set of four digits at the start of every vial label. They should all be identical and formatted as "HJW_####." **Make sure to add an identifier for other samples collected at the same location.**
2. Record the general weather conditions, river intermittency, vegetation type, sediment type, hydrogeomorphology, and coverages of algae, in-stream plants, and overhead canopy using the checklists/categories on the metadata form.
3. Record the latitude and longitude of the **sensor deployment** location in **decimal degrees**. You can use a GPS or a smartphone app such as 'My GPS Coordinates' or Google maps. **If no coordinates are available due to poor signal, write down the site name.**
4. Measure the water depth where sediments will be collected and record it on the datasheet in centimeters.
5. With a smartphone or camera held horizontally, take a photo looking straight down through the water of the undisturbed sediment so that about 1m^2 can be seen. Hold the camera/phone **parallel** to the riverbed so that the image is taken looking straight down to avoid distortion. If possible, place a measuring tape extended to 30 cm on the bed of the river so it is visible in the photo. Note that this will be uploaded along with the metadata.
6. In addition, take the following photos (note these will be uploaded along with metadata). For the last three pictures (b, c, d), lay out a **measuring tape to 30cm** as a reference for scale and make sure it can be seen in the photos. The first picture (looking across the river) doesn't need to contain the measuring tape. Try to hold the phone/camera in a **horizontal** orientation if possible such that the picture should be wider than it is tall. Please avoid taking pictures with people in them.
    a. Looking across the river to give a sense of how broad the river is
    b. Looking upstream, showing the river surface, shoreline sediments, and vegetation
    c. Looking downstream, showing the river surface, shoreline sediments, and vegetation
    d. Looking straight down at exposed shoreline sediments (not under water) so that about 1m^2 can be seen in the image. Hold the camera **parallel** to the sediments to minimize distortion. (If the sediment is sloped, hold the camera at a matching slope to keep it

parallel.)


14. If you have a device for measuring temperature, DO, Spc or pH; record it on the metadata sheet in degrees C.
15. Measure the height of the water column at the place you plan to sample by extending a measuring tape from the riverbed vertically to the water surface. Record on the metadata sheet in centimeters.
16. Please confirm that all fields on the paper metadata sheet are filled out prior to leaving the field, and take a picture of the data sheet. This picture will also be uploaded. You will fill in the rest of the metadata via an online form. You will need to fill out a few fields after collecting the samples.

COLLECT FILTERED WATER SAMPLES

Follow the instructions below to collect your water samples from the water column above the depositional zone you will collect sediment from. Collect all the water samples before beginning sediment collection.

1. Put on nitrile gloves.
2. Locate all the labeled materials (vials).
3. After collection, vials should go into a cooler with ice.

SAMPLE PHASE 1: FILTERED WATER SAMPLES - Three pre-acidified 60mL AMBER GLASS vials (Wear safety glasses) *These vials are for FTICR-MS. The label suffix after the sample ID is "ICR" and a replicate number (-1, -2, -3)*

4. Locate the three 60mL amber vials labeled as "ICR". These contain a very small volume of hydrochloric **acid** (safety glasses recommended).
5. Also locate the filter.
6. Open the 50 mL syringe package and remove the syringe. Please **don't touch the outlet of the syringe and do not fully remove the plunger from the syringe body**. This is important to avoid contamination.
7. While sampling, please **stand downstream of the sampling location and point the opening of the syringe upstream**. This is important to collect a representative sample and to avoid getting disturbed sediment into the water sample.
8. Fill the syringe with river water, collecting water from 50% of the water column depth. Expel the syringe contents into the river (downstream of the sampling location) and repeat this two more times. You only need to flush the syringe as described when you first open the syringe package. You do not need to flush repeatedly when collecting additional syringe volumes.
9. After flushing the syringe 3 times, fill the syringe again from 50% of the water column depth; this is the sample water to be collected.
10. Screw the syringe onto the filter. Please **don't touch the outlet of the syringe, or the inlet of the filter**. This is important to avoid contamination.
11. Push 5 mL of water through the filter/needle assembly. This water is not collected.
12. Open the 60mL amber vial and **without touching the top of the vial**, filter water into the acidified **amber** glass vials **to the premarked line**. Then re-cap the vial (don't let anything touch the inside of the vial or the inside of the cap, to avoid contamination).
13. Shake the vial gently to incorporate the acid into the sample. Store vial on ice.
14. Prior to collecting the second replicate, remove the filter from the syringe, and push out all the water, then refill the syringe and push at least 5ml of water through the filter (this is to push new sample into the filter housing).
15. Collect the second replicate as described above into the second vial, then refill the syringe, flush the filter, and collect the third replicate into the third vial. Store vials on ice.

SAMPLE PHASE 2: FILTERED WATER SAMPLES - Three 40mL AMBER GLASS vials *These vials are for DOC and total dissolved N. The label suffix after the kit ID is "OCN" and a replicate number (-1, -2, -3)*

19. Locate the 40mL amber vials labeled as "OCN".
20. Collect water into the syringe as described above and flush 5 ml through the filter.
21. Open the vial and fill one of the 40ml glass vials so it is ~ ¾ full by holding the filter outlet above the vial opening. Then re-cap the vial (don't let anything touch the inside of the vial or the inside of the cap, to avoid contamination).
22. Refill the syringe as described above, flush 5 ml through the filter and collect the second replicate in the second vial, and repeat for the third replicate in the third vial. Store vials on ice.

SAMPLE PHASE 3: COLLECT BULK SEDIMENT

1. Locate the sediment materials set aside earlier. These should be:
    a. Three 50ml white-capped conical tubes marked as "SED".
    b. Clean sieve and catch pan
2. Locate the foil-wrapped scoop and scoopula. You can use the scoopula to remove plant material and larger (>4mm) debris and help get sediment off the scoop and into the container
3. Put on a new pair of nitrile gloves to be used for sediment collection.
4. **Never scrape the sides of the plastic vials with the sediments or tools as this could introduce plastic contamination. If needed, use the scoop and scoopula in tandem to drop sediment down from the opening.**
5. Remove the 30mL metal scoop from packaging; ensure you only touch the scoop handle and not the spoon. Wash the scoop, scoopula, sieve and catch pan by dipping them in the stream. If you need to set down the scoop or scoopula, place them on the clean side of the foil to minimize contamination.
6. Use the scoop to collect **underwater** surface sediments (**1-3 cm depth**, ideally, but can go a little deeper if needed) from the streambed. The goal is to fill each conical tube with ~15 mL of sieved sediment with as little water as possible. To do this, decant any bulk water. Sampling should avoid plant material and avoid sediments > 4 mm so there is enough < 2mm sediment for analysis.
7. Sieve sediment into the catch pan and then use the metal tools to scoop ~15 mL of sieved sediment from the catch pan into each conical tube. **Do not use your hands (even if gloved) to push material through the sieve or scoop the sediment.**
7. To get enough material, it is okay to move around and collect sediments from across the depositional zone and even to move to additional nearby depositional zones with a similar physical setting and similar sediments.
8. The most important thing is getting enough fine-grained sediment.
9. Store tubes with sediment in the cooler.
10. Fill out the metadata sheet field that asks about sediment collection depth.